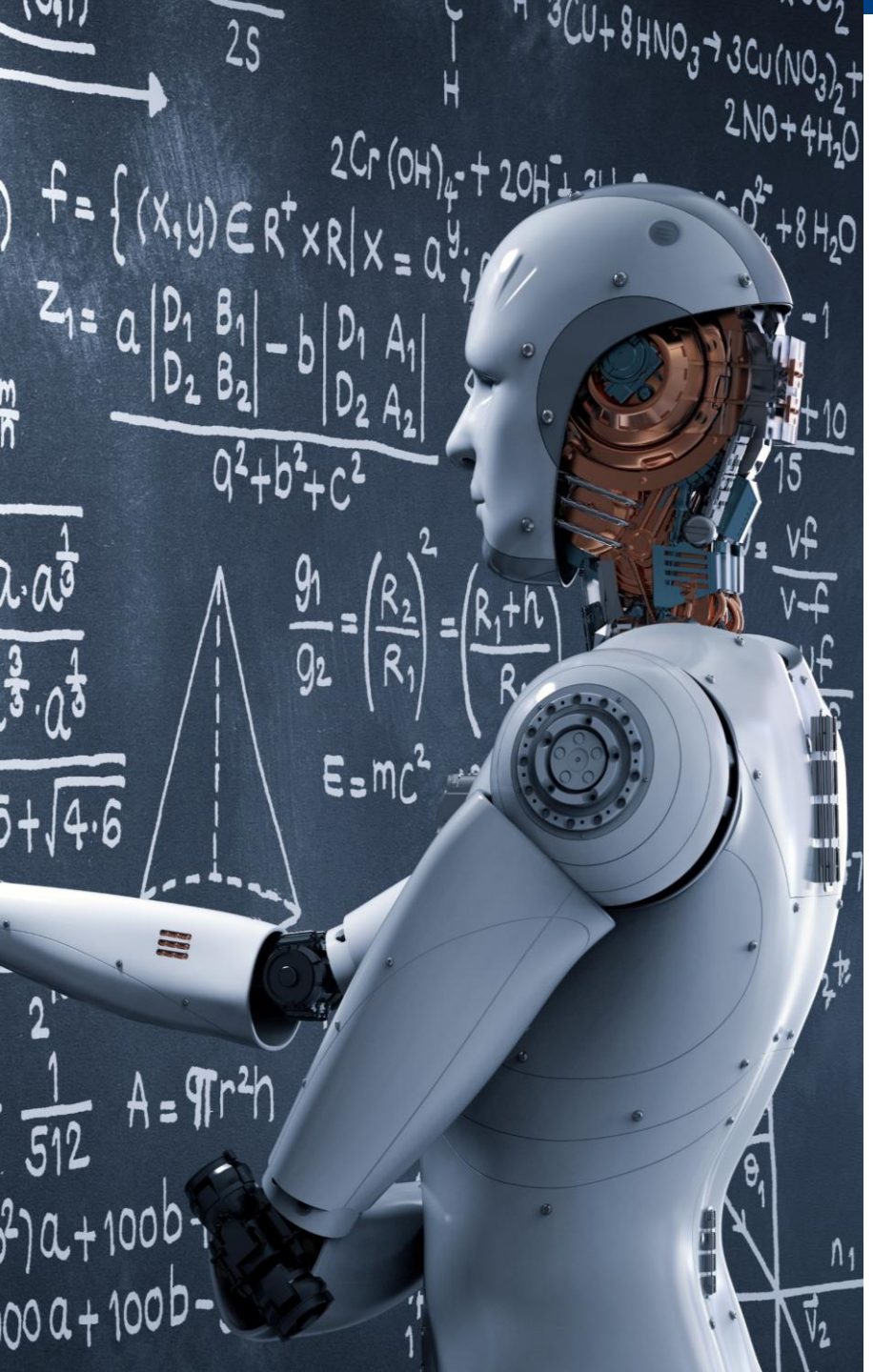

# Session 1: Welcome and introduction

## Managing AI-based Systems

**Prof. Dr. Nils Urbach**

Frankfurt University of Applied Sciences,
Research Lab for Digital Innovation & Transformation

FIM Forschungsinstitut für Informationsmanagement

Fraunhofer-Institut für Angewandte Informationstechnik FIT,
Institutsteil Wirtschaftsinformatik

www.ditlab.org
www.fim-rc.de
www.wirtschaftsinformatik.fraunhofer.de

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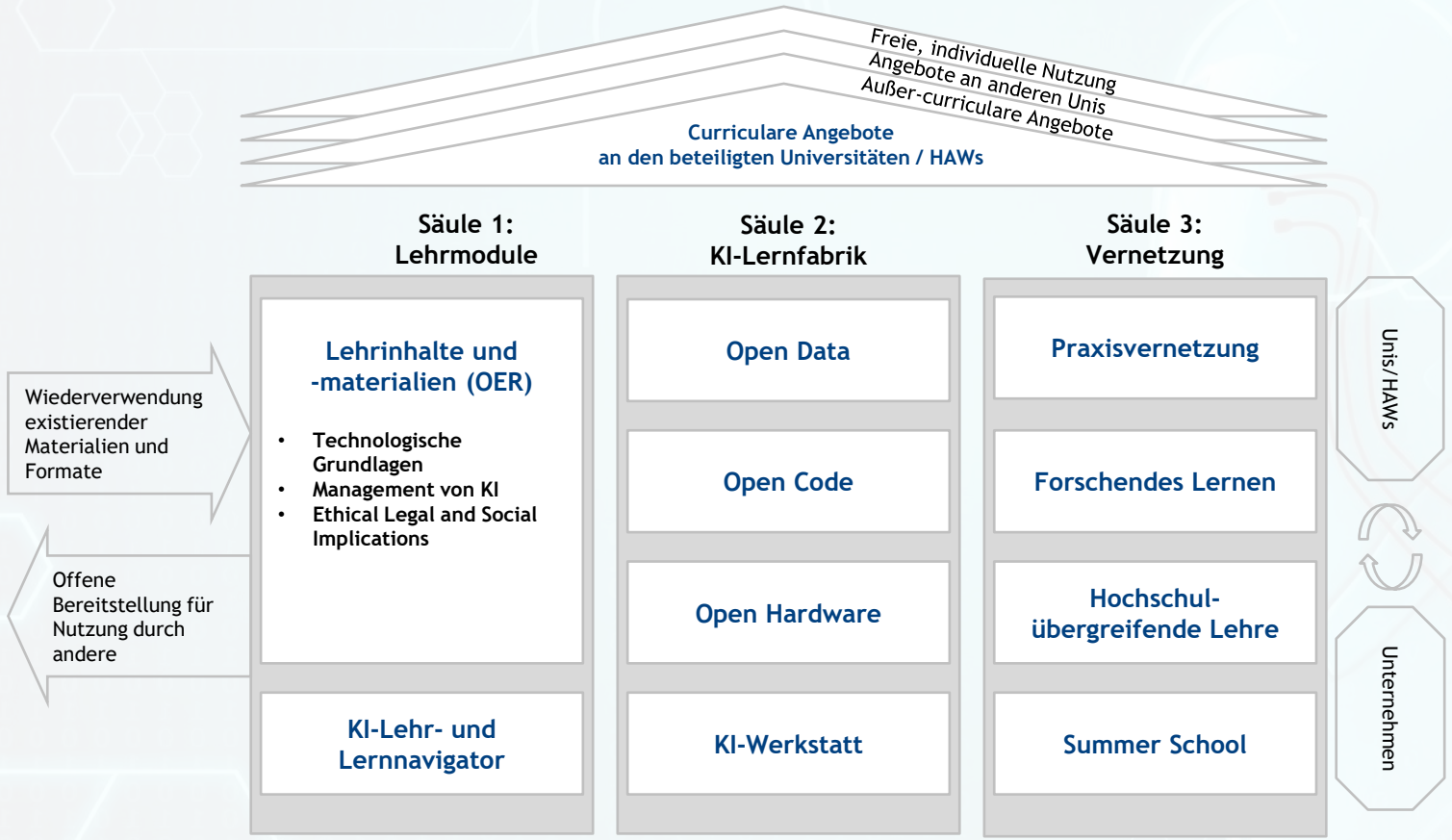
# ABBA - AI for Business | Business for AI




AI FOR BUSINESS
BUSINESS FOR AI

gefördert durch:

Bundesministerium für Bildung und Forschung

HESSEN
Hessisches Ministerium für Wissenschaft und Kunst

Baden-Württemberg
MINISTERIUM FÜR WISSENSCHAFT, FORSCHUNG UND KUNST

Projektpartner:

FRANKFURT UNIVERSITY OF APPLIED SCIENCES

UNIVERSITÄT HOHENHEIM

UNIVERSITÄT BAYREUTH

KIT
Karlsruher Institut für Technologie

# Current reference to motivate taking part in our lecture

Künstliche Intelligenz

**Diese KI-Führungskräfte brauchen Unternehmen jetzt**

Ingenieure, Evangelisten, Regenmacher: Um KI erfolgreich einzusetzen, brauchen Unternehmen neue Managementrollen. Die fünf wichtigsten Typen und wie Unternehmen sie richtig einsetzen.

Lina Knees
07.10.2023 - 10:05 Uhr

- **AI Relation Manager:**
  - especially for companies important that already implemented AI
  - act as a coordinator between technology and business
  - know the product, the employees and supplier and build an interface between them
  - have a balanced mix of technical knowledge, business acumen and social skills
- **AI Rainmaker:**
  - enable companies that want to maximize their business with AI
  - scrutinizes processes, identifies problems that can be solved with AI and develops a long-term strategy for the use of AI
  - scientific background, a good understanding of business and many years of experience in strategy or transformation projects
  - confident appearance, a clear vision and the courage to dare something new
- **AI Engineer:** not part of this lecture

This lecture will train you to become AI managers – a highly demanded job position in the industry now and in the future

Knees, 2023

# Current reference to motivate taking part in our lecture

Künstliche Intelligenz

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Lina Knees
07.10.2023 - 10:05 Uhr

- **AI Evangelist:** not part of this lecture
- **AI Ethicist:**
  - advise companies on the impact of AI, taking into account the interests of government, employees and society
  - balance economic and technical expertise with sociological perspectives
  - Establish processes to continuously review and assess AI systems for ethical risk, for example, to prevent misinformation or misconduct by AI

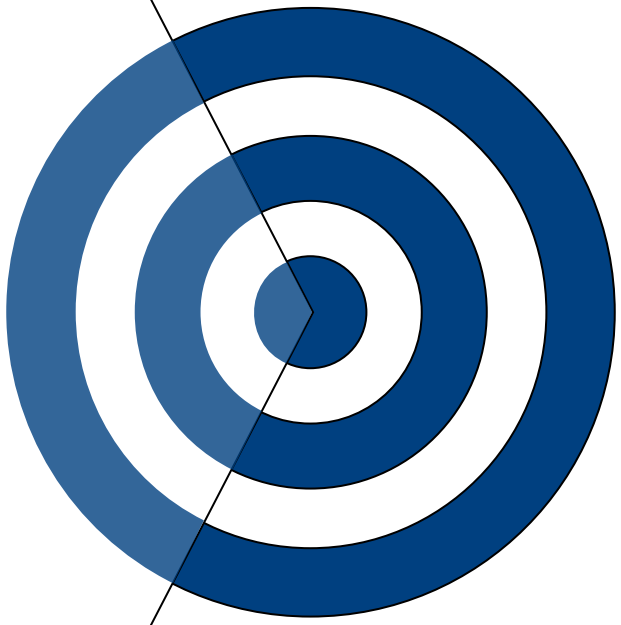
**Overall, this course will give you unique knowledge to become one of the AI managers - use this chance and actively participate in the course!**

This lecture will train you to become AI managers – a highly demanded job position in the industry now and in the future

Knees, 2023

# Objectives of today's lecture

1. Define and delineate Artificial Intelligence (AI)
2. Comprehend the historical development of AI
3. Understand the hype around AI
4. Obtain an administrative overview of the entire course



01 Motivation

02 Definition

03 AI history

04 Why now?

05 Course outlook

06 Administrative topics

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02 Definition

03 AI history

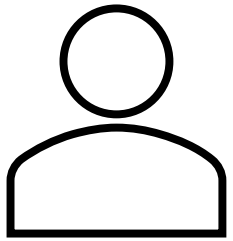
04 Why now?

05 Course outlook

06 Administrative topics

# Human or Artificial Intelligence

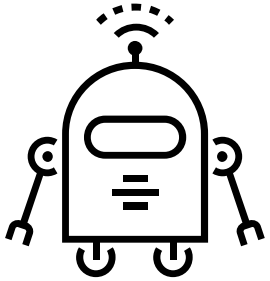
http://www.bbc.com/news/technology-35977315



Human

**A**

**or**



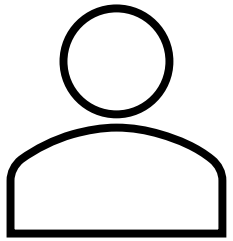
AI

**B**

(1) nextrembrandt.com

# Human or Artificial Intelligence

http://www.bbc.com/news/technology-35977315



Human

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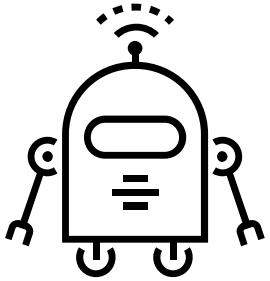

AI

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(1) nextrembrandt.com

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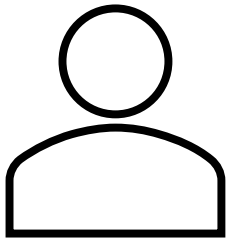

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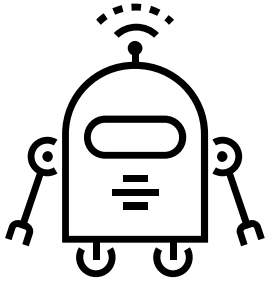

AI

**B**

(1) theverge.com, 2019

# Human or Artificial Intelligence

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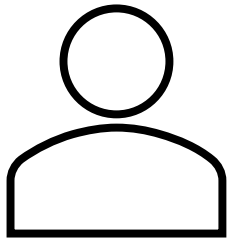

Human

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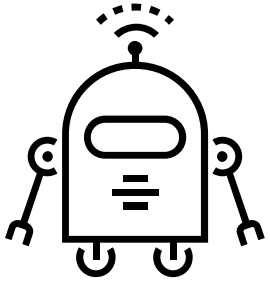


AI

**B**

(1) unsplash.com

# DALL-E conquers the world

News

**Dall-E Mini: Diese KI kreiert faszinierende und zugleich gruselige Bilder**

**DALL E-2 CAN CREATE ITS OWN SECRET CODE! IS IT COMMUNICATING WITH ALIENS?**

**Dall.E mini: The AI image generator everyone's using to make wild memes**

Comment

/ Live news

**Is AI the future of art?**

Kreativer Algorithmus DALL.E

**Diese KI macht aus jedem Text ein Bild**

**Kunstwerk auf Kommando: KI den Pinsel schwingt**

Neuronale Netzwerke erzeugen zunehmend bessere Bilder nach

**„Unheimlich": Diese KI erstellt Bilder aus Text**

# DALL·E 2 – Art of tomorrow?

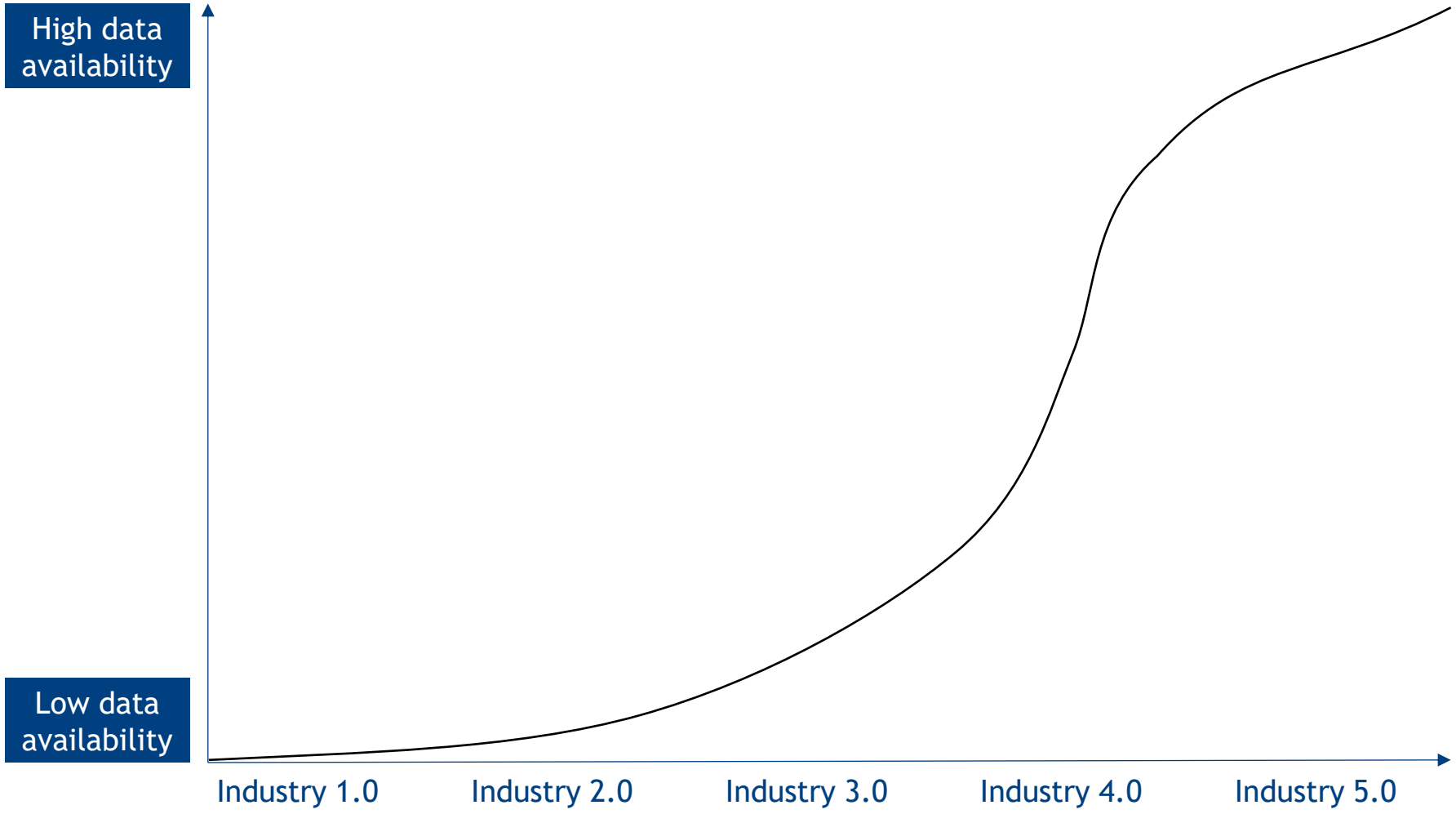
a dolphin in an astronaut suit on saturn, artstation

a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese

a teddy bear on a skateboard in times square

(1) Ramesh et al. (2022)

# Digital Trends




Birth of **Industry 5.0:**

Utilization of Big Data and the Internet of Things (IoT) with **artificial intelligence (AI)**

(1) Özdemir & Hekim, 2018, (2) twi-global.com

# Companies with digital business models overtake established players with classic business models

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## Largest companies in the world by market value

**2009** (third quarter)

| | |
|---|---|
| 1. Exxon Mobil | 329.725 |
| 2. PetroChina | 325.097 |
| 3. ICBC | 237.951 |
| 4. Microsoft | 229.630 |
| 5. HSBC | 198.561 |
| 6. China Mobile | 195.680 |
| 7. Wal-Mart | 189.331 |

**2019** (third quarter)

| | |
|---|---|
| 1. Microsoft | 1.062.000 |
| 2. Apple | 1.012.000 |
| 3. Amazon.com | 858.680 |
| 4. Alphabet | 838.020 |
| 5. Berkshire Hathaway | 508.530 |
| 6. Facebook | 508.050 |
| 7. Alibaba Group | 435.400 |

All market capitalisation figures are in millions of USD.

“The world’s most valuable resource is no longer oil, but data.”

*- The Economist 2017*

„If data is the new oil of our economy […], then AI can analogously be referred to as the engine.“

*- Agrawal et al. 2018*

# Introducing AI often presents companies with several challenges

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**Employees need to accept and build trust in AI applications**

**New requirements for company's workforce, with a shift towards hiring AI specialists**

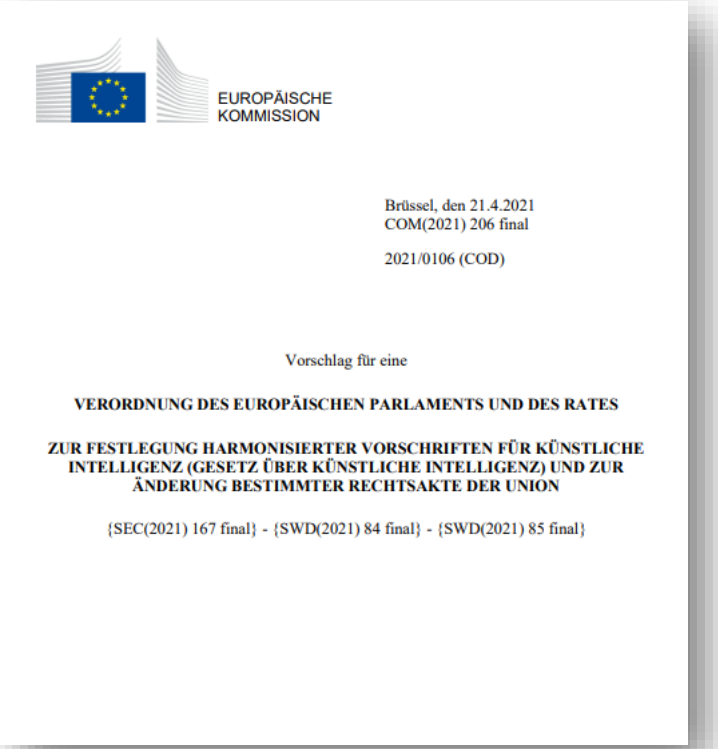
**Data quality, availability, and correct usage challenges**

**Data privacy and data loss challenges**

**Liability question**

AI brings significant changes: At the business level, processes and tasks are greatly impacted using AI

# A trend towards more deployment of AI – The EU AI Act regulates the development and use of AI within the EU



EUROPÄISCHE KOMMISSION

Brüssel, den 21.4.2021
COM(2021) 206 final

2021/0106 (COD)

Vorschlag für eine

**VERORDNUNG DES EUROPÄISCHEN PARLAMENTS UND DES RATES**

**ZUR FESTLEGUNG HARMONISIERTER VORSCHRIFTEN FÜR KÜNSTLICHE INTELLIGENZ (GESETZ ÜBER KÜNSTLICHE INTELLIGENZ) UND ZUR ÄNDERUNG BESTIMMTER RECHTSAKTE DER UNION**

{SEC(2021) 167 final} - {SWD(2021) 84 final} - {SWD(2021) 85 final}

## Target of the EU regulation

1. It must be ensured that the AI systems launched and used on the Union market and used are **safe** and **respect the existing fundamental rights and values** of the European Union.
2. To promote investment in AI and innovative AI, **legal certainty** must be guaranteed.
3. **Governance** and the effective enforcement of applicable law to safeguard fundamental rights as well as the security requirements for AI systems must be strengthened.
4. The **development of a domestic market** for legally compliant, secure, and trustworthy AI applications must be facilitated and market fragmentation must be prevented.

**There is a need for AI governance and AI management**

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Implementation

01 Motivation

02 Definition

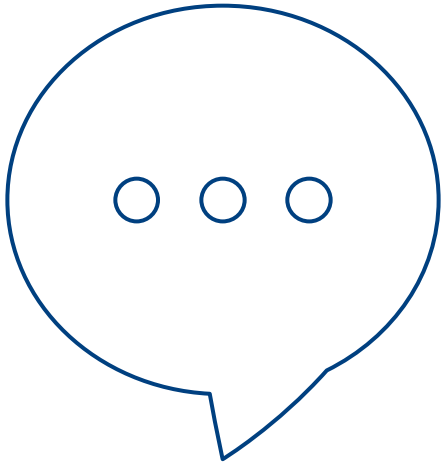
03 AI history

04 Why now?

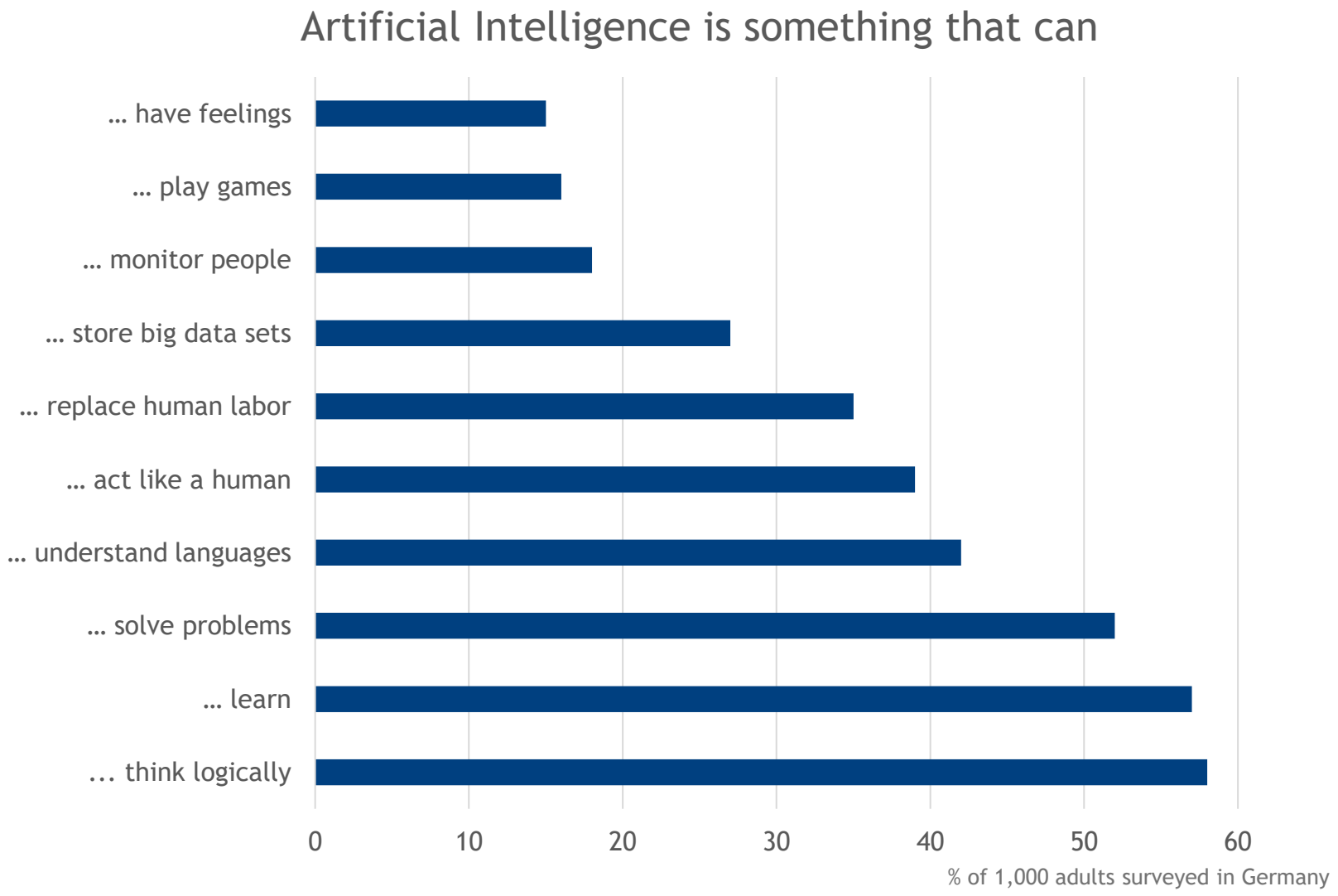
05 Course outlook

06 Administrative topics

# What do you think of when you think of AI?



# What do 1,000 people think of when they think of AI?




(1) Pegasystems, (2) Statista

# Definitions – Artificial Intelligence

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"The goal of AI is to develop machines that behave as if they possess intelligence" (McCarthy, 1956)

"Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better" (Rich & Knight, 1991)

"Artificial intelligence (AI) refers to systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve specific goals" (European Commission, 2018, p. 1)

"Artificial intelligence is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings" (Copeland, 2023)

**There is no single all-encompassing definition of AI**

# What is AI in this lecture?

There is not the one AI, but there are various AI technologies, applications, perspectives, and research fields

### Philosophy

- Knowledge
- Rules
- Validity

### Mathematics

- Computing
- Formalization
- Logic
- Probability

### Economics

- Profit
- Utility
- Decision theory

### Neuroscience

- Information processing of the brain

### Psychology

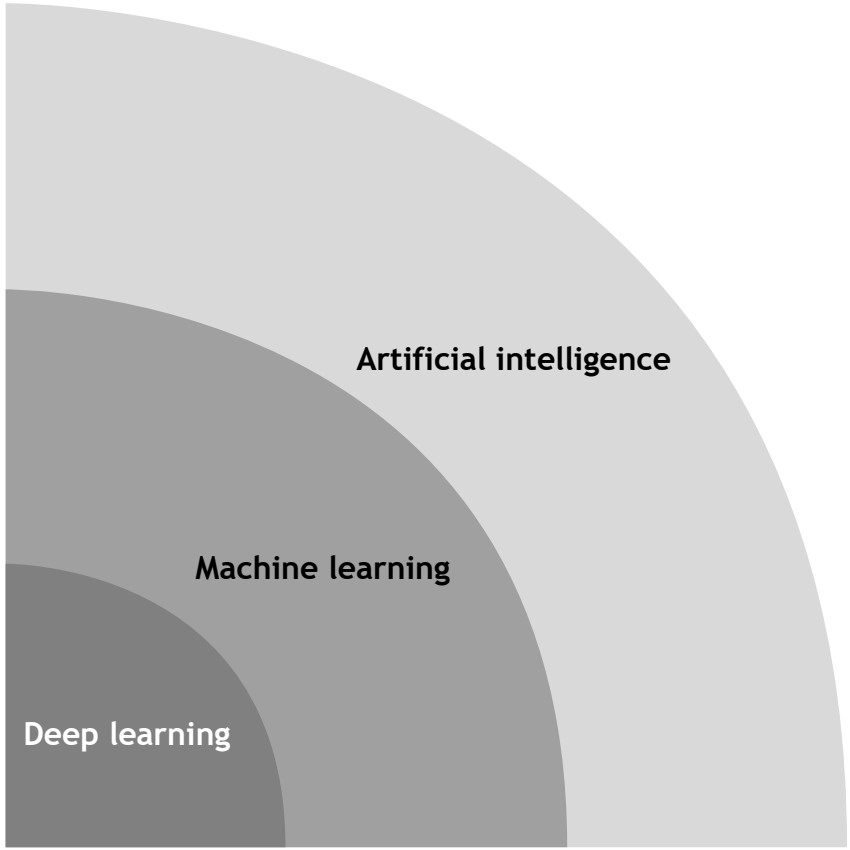
- Human behavior
- Cognitive psychology

### Computer engineering

- Programming
- Operational systems

(1) Russel und Norvig, 2010

# AI vs machine learning vs deep learning




**Artificial Intelligence** can be based on programmed processes or generated by machine learning or deep learning algorithms

**Machine learning** are methods of learning processes that can be used to identify relationships/ patterns in existing data sets in order to make predictions based on them

**Deep learning** is a specialized branch of machine learning that employs artificial neural networks with multiple layers to process and learn from extensive data

(1) Murphy, 2012

01 Motivation

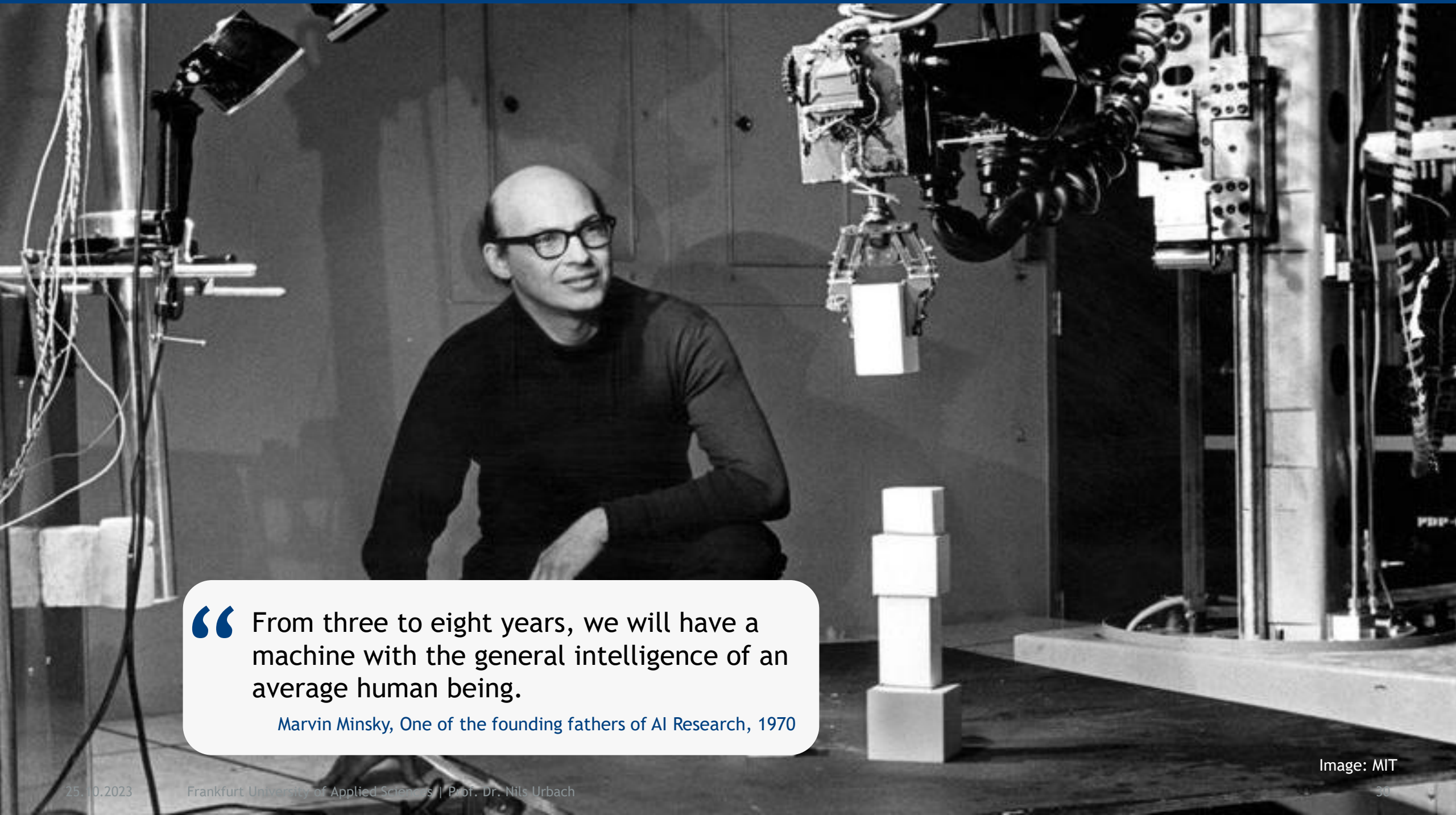
02 Definition

03 AI history

04 Why now?

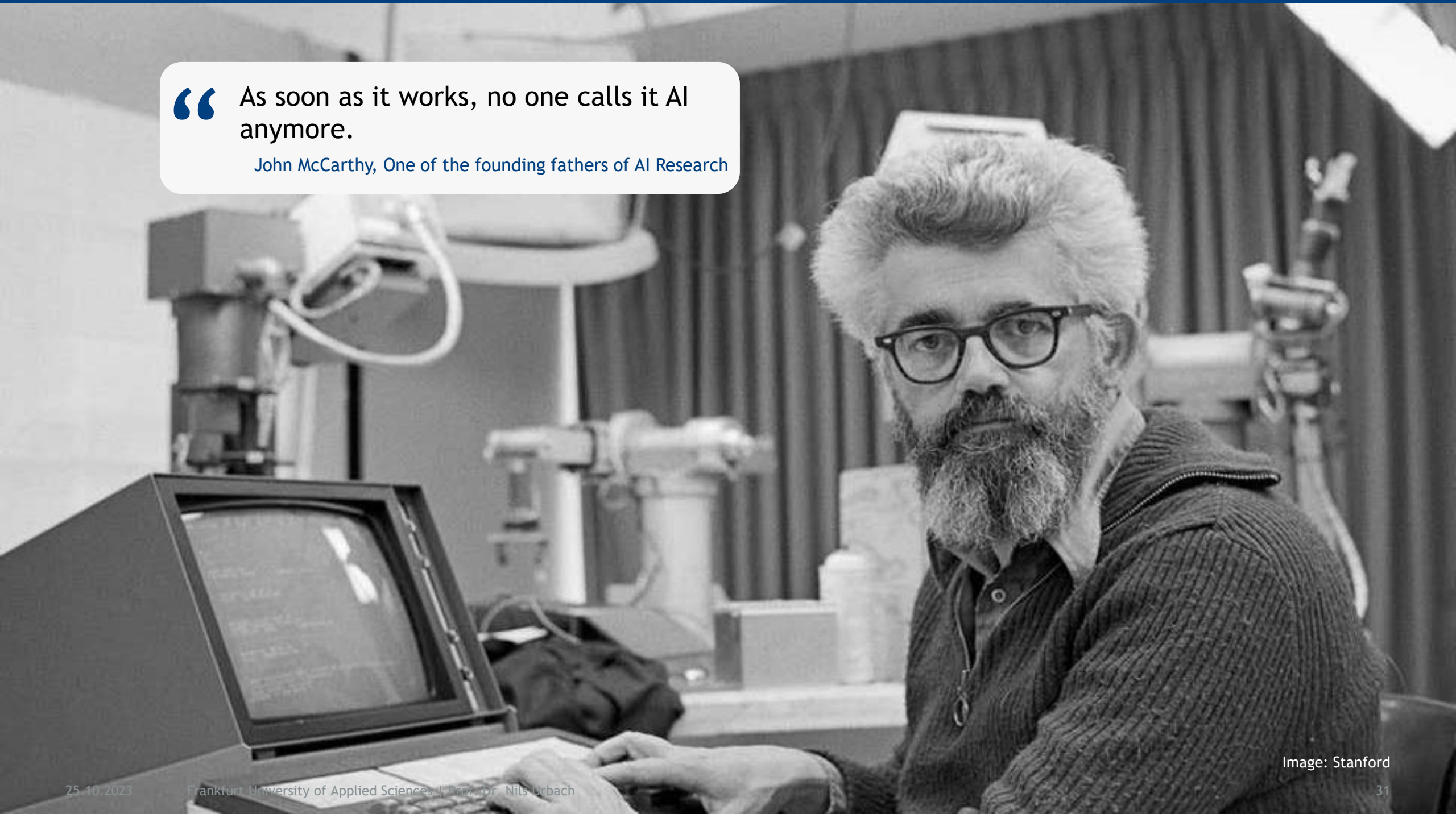
05 Course outlook

06 Administrative topics



> From three to eight years, we will have a machine with the general intelligence of an average human being.
>
> Marvin Minsky, One of the founding fathers of AI Research, 1970

Image: MIT



> As soon as it works, no one calls it AI anymore.
>
> John McCarthy, One of the founding fathers of AI Research

Image: Stanford

# The long history of Artificial Intelligence

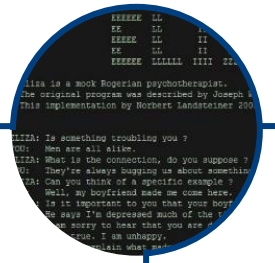
**1950**

**Turing Test**

**1966**

**ELIZA**

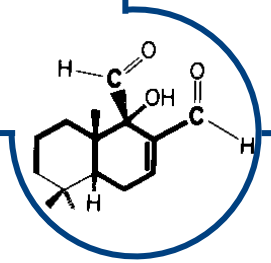
First Chatbot



**1972**

**MYCIN**

Expert system



**1978**

**R1**

Expert system

(1) Wikimedia, (2) Wikipedia, (3) Jeopardy, (4) Wired, (5) Forbes, (6) Zeit online

# The long history of Artificial Intelligence

## 1997

**Deep Blue**

wins chess match

## 2011

**IBM Watson**

wins Jeopardy

## 2016

**Alpha GO**

wins GO

## 2020

**BlueDot**

AI warns the world about covid-19

(1) Wikimedia, (2) Wikipedia, (3) Jeopardy, (4) Wired, (5) Forbes, (6) Zeit online

# AI in action in the Covid-19 pandemic

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## AI is instrumentalized in the fight against the Corona virus

### Support in the search for a vaccine

- AI examines compounds and matches them with a parameter database
- Shortening the period until the first tests of the vaccine

### Prediction of the pandemic

- Early detection of the virus, through analysis of data sets such as news, airline ticket sales, demographic data, and climate data
- Forecasting Corona hotspots

### Quick diagnoses

- AI exchanges data with CT scanners in hospitals
- AI analyzes CT images of the lungs, identifies the signs of coronavirus and evaluates the changes

**In addition, AI supports the nursing staff, controls citizens, evaluates policies and measurements, ...**

01 Motivation

02 Definition

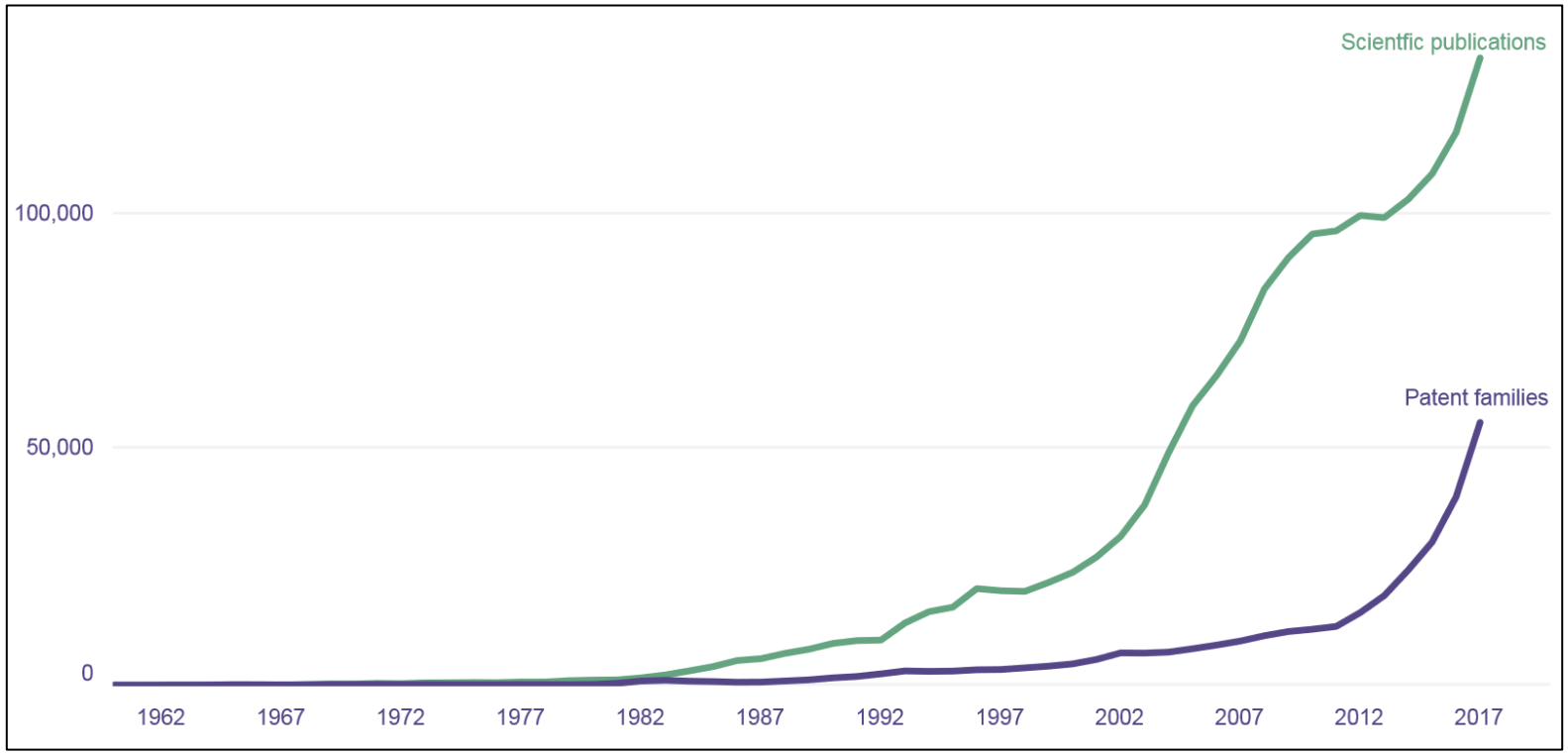
03 AI history

04 Why now?

05 Course outlook

06 Administrative topics

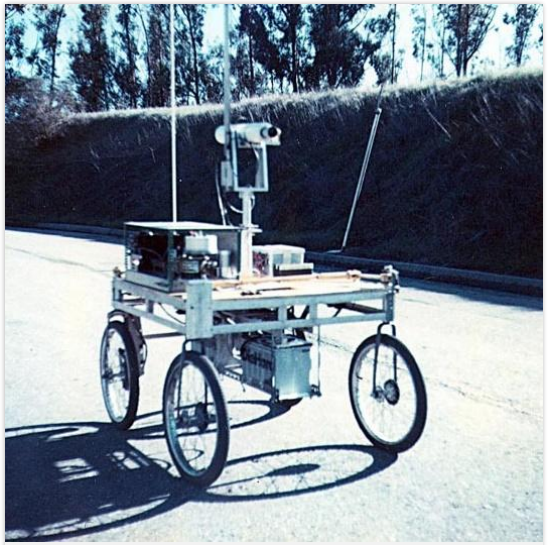
# Artificial Intelligence - Why now?




Between 1960 and early 2018, nearly 340,000 patent families and more than 1.6 million scientific papers related to Artificial Intelligence were published

# Artificial Intelligence - Why now?

Past



Stanford Cart (1971)

Present

Tesla Autopilot (2018)

In the past, the focus was particularly on basic research in the field of AI. Today, however, AI is increasingly being used in business models, services and products.

(1) Stanford, (2) Tesla

# Why now?
# Four key developments enable the application of AI

### Necessary data is available and usable

Digital technologies create data that can be used for AI applications through Big Data technologies

### The machine learning algorithms have been improved

In recent years, significant progress has been made, especially through deep learning algorithms

### Cloud services provide the required computing power

Cloud services enable fast, flexible and affordable use of computing resources without major upfront investments

### The application of AI is no rocket science

High-performance (open source) toolkits and libraries are available

01 Motivation

02 Definition

03 AI history

04 Why now?

05 Course outlook

06 Administrative topics

# AI trends and developments provide major opportunities and risks for businesses

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## AI trends

- Data analytics & management
- Machine learning & RPA
- Autonomous driving
- Cybersecurity
- Creative AI & Metaverse

## Opportunities and risks

- \+ Increased productivity
- \+ Increased efficiency
- \+ Increased flexibility and speed
- \+ Improved ROI
- – Flawed data
- – Implementation errors
- – Security risks
- – Blind faith in AI

**There is a need to manage AI systems (e.g., risk management, data management, …)**

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Implementation

**L2: Technical & Mathematical Fundamentals**
- Technical basics
- AI learning methods
- Mathematical algorithms

**L3: Neural Networks**
- Basic intuition
- Training process of neural networks
- Large language models

**L4: AI Metrics**
- AI models
- Metrics
- Hyperparameter optimization

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8

**L5: Application potentials of AI technologies**
- Human AI interaction
- AI-based process optimization
- Cross-technology AI innovation

**L6: Identification and evaluation of AI use cases**
- Identifying AI use cases
- Designing AI use cases
- Evaluating AI use cases

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**AI Operations at Scale**

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**L7: AI readiness**
- Organizational AI readiness factors
- Integrating AI service platforms

**L8: Management and governance for AI**
- AI application management model
- Team structures
- AI Governance

**ns at Scale**

Monitoring
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Implementation

# The AI implementation process consists of four phases

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4

AI Ide

5 Application p techno

6 Identificatio evaluation o

**L9: Architectures of AI applications**
- ML decision space
- Data, training and deployment infrastructure
- From DevOps to MlOps

**L10: Data Management and Model Transparency**
- Data literacy
- Model lineage
- Data quality management

**L11: Design of Human-AI interaction**
- Transparency & explainability
- AI fairness
- Delegation

**AI Design & Development**

9 Architectures of AI applications

10 Data Management and Model Transparency

11 Design of human-AI interactions

**AI Operations at Scale**

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# The AI implementation process consists of four phases

2
3 Basic understanding of AI and its business potential
4

AI Ideation

5 Application potentials of AI technologies

6 Identification, design and evaluation of AI use cases

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**L12: KPI-based Monitoring and Change Management**
- Key performance indicators for AI
- Machine learning monitoring

**L13: Ethical, legal and social implications of AI**
- Moral agency
- Trustworthy AI
- Legal implications

AI Operations at Scale

12 KPI-based Monitoring and Change Management

13 Ethical, legal and social implications of AI

Implementation

# Today's lecture at a glance

- We understand AI as systems that display intelligent behavior by analyzing their environment and taking actions – with some degree of autonomy – to achieve specific goals
- We know that AI is not the perfect solution for everything and thus has chances and risks
- We understand the need for systematic management of AI applications
- We know and understand the administrative framework of the course

# Questions, comments, observations

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# Pictures

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